MA1522 Linear Algebra for Computing

Lesson 12: Row Space, Column Space, Nullspace and Rank

Regarding citing theorems in final exam:

Good Example:

Since $\text{rank}(A)=2$, $\text{rank}(B)=3$, by the ineq $\text{rank}(AB) \le \min\{\text{rank}(A), \text{rank}(B)\}$, we conclude that $\text{rank}(AB) \le 2$

Bad example

1. Since - - - -, by thm XXX in the textbook, $\text{rank}(AB) \le 2$. ×

2. Since, $\text{rank}(A)=2$, $\text{rank}(B)=3$, we have $\text{rank}(AB) \le 2$. not very good.

3. By the above comparison, $\text{rank}(AB) \le 2$. too rough, no reason is showing. ×

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson12` (e.g. `A1234567Z_lesson12.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 12**.
- Deadline: today, **10:30 AM**.
- If you do not know how to upload your submission, come to ask me after class.

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Section 4.1 Column Space, Row Space, and Nullspace

Let A be an $m \times n$ matrix, written by columns and by rows as

$$A = \begin{pmatrix} | & | & & | \\ \mathbf{c}_1 & \mathbf{c}_2 & \cdots & \mathbf{c}_n \\ | & | & & | \end{pmatrix} = \begin{pmatrix} — & \mathbf{r}_1 & — \\ & \vdots & \\ — & \mathbf{r}_m & — \end{pmatrix}.$$

(m rows, n columns)

Definition

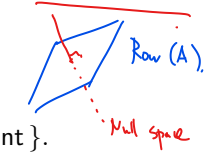
1. **Row space**: $\text{Row}(A) = \text{span}\{\mathbf{r}_1, \ldots, \mathbf{r}_m\} \subseteq \mathbb{R}^n$.
2. **Column space**: $\text{Col}(A) = \text{span}\{\mathbf{c}_1, \ldots, \mathbf{c}_n\} \subseteq \mathbb{R}^m$.
3. **Nullspace**: $\text{Null}(A) = \{\, \mathbf{v} \in \mathbb{R}^n \mid A\mathbf{v} = \mathbf{0} \,\} \subseteq \mathbb{R}^n$, and $\text{nullity}(A) = \dim(\text{Null}(A))$.

Remark.

Null space is the orthogonal complement of row space:

$$Av = \begin{pmatrix} — & r_1 & — \\ & \vdots & \\ — & r_m & — \end{pmatrix} \begin{pmatrix} | \\ v \\ | \end{pmatrix} = \begin{pmatrix} r_1 \cdot v \\ r_2 \cdot v \\ \vdots \\ r_m \cdot v \end{pmatrix}$$

So $Av = 0 \iff r_i \cdot v = 0 \ \forall i$.



Column space and consistency

$$\text{Col}(A) = \{\, A\mathbf{u} : \mathbf{u} \in \mathbb{R}^n \,\} = \{\, \mathbf{b} : A\mathbf{x} = \mathbf{b} \text{ is consistent} \,\}.$$

Section 4.1 Bases from the RREF

Suppose

$$A = \begin{pmatrix} | & & | \\ \mathbf{c}_1 & \cdots & \mathbf{c}_n \\ | & & | \end{pmatrix} = \begin{pmatrix} — & \mathbf{r}_1 & — \\ & \vdots & \\ — & \mathbf{r}_m & — \end{pmatrix} \xrightarrow{\text{rref}} R = \begin{pmatrix} | & & | \\ \mathbf{c}'_1 & \cdots & \mathbf{c}'_n \\ | & & | \end{pmatrix} = \begin{pmatrix} — & \mathbf{r}'_1 & — \\ & \vdots & \\ — & \mathbf{r}'_m & — \end{pmatrix}$$

Theorem

1. Row operations preserve row space: $\text{Row}(A) = \text{Row}(R)$. Exercise: check this
2. Row operations preserve **linear relations between columns**: $x_1\mathbf{c}_1 + \cdots + x_n\mathbf{c}_n = \mathbf{0}$ if and only if $x_1\mathbf{c}'_1 + \cdots + x_n\mathbf{c}'_n = \mathbf{0}$.

As a consequence, the subspaces $\text{Row}(A)$, $\text{Col}(A)$ and $\text{Null}(A)$ can be computed as follows.

(a) the nonzero $\mathbf{r}'_i$ form a basis for $\text{Row}(A)$;

Exercise: In an RREF, {non-pivot columns} $\subseteq$ span({pivot columns})

(b) if $\mathbf{c}'_{i_1}, \ldots, \mathbf{c}'_{i_r}$ are the pivot columns of R, then $\mathbf{c}_{i_1}, \ldots, \mathbf{c}_{i_r}$ form a basis for $\text{Col}(A)$;

(c) $\text{Null}(A)$ is read off from the general solution of $R\mathbf{x} = \mathbf{0}$.

Question 1

Warning: Row operation does not preserve col(A).

eg. $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \xrightarrow{R_2 - R_1} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} = B$

$\text{col}(A) = \text{span}\left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right\} \neq \text{col}(B) = \text{span}\left(\left\{\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right\}\right)$

Let $A = (\mathbf{a}_1\ \mathbf{a}_2\ \mathbf{a}_3\ \mathbf{a}_4\ \mathbf{a}_5)$, where

$$A = \begin{pmatrix} 1 & -2 & 1 & 2 & 0 \\ 2 & -4 & 3 & 3 & 0 \\ 1 & -2 & 2 & 1 & 1 \\ -1 & 2 & 1 & -4 & -1 \end{pmatrix} \xrightarrow{\text{rref}} R = \begin{pmatrix} 1 & -2 & 0 & 3 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

(a) Find a basis for $\text{Row}(A)$.

(b) Find a basis for $\text{Col}(A)$, using columns of A.

(c) Find a basis for $\text{Null}(A)$.

Question 1 — solution

$A = \begin{pmatrix} 1 & -2 & 1 & 2 & 0 \\ 2 & -4 & 3 & 3 & 0 \\ 1 & -2 & 2 & 1 & 1 \\ -1 & 2 & 1 & -4 & -1 \end{pmatrix} \xrightarrow{\text{rref}} R = \begin{pmatrix} 1 & -2 & 0 & 3 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$

$c_1' \; c_2' \; c_3' \; c_4' \; c_5'$ — pivots: c_1', c_3', c_5'. These rows are 0.

$\in \mathbb{R}^5$.

1. $\text{Row}(A) = \text{Row}(R) = \text{span}\left\{ \begin{pmatrix} 1 & -2 & 0 & 3 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 & -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \end{pmatrix} \right\}$.

this is a basis for Row(A).

$\begin{cases} c_2' = -2c_1' \\ c_4' = 3c_1' + (-1)\cdot c_3' \end{cases}$.

2. $\text{Col}(A)$: Since $\{c_1', c_3', c_5'\}$ is a basis of $\text{col}(R)$.

we have $\{a_1, a_3, a_5\}$ forms a basis of $\text{col}(A)$.

$\begin{cases} a_2 = -2a_1 \\ a_4 = 3a_1 + (-1)a_3 \end{cases}$

(You can check this).

3. $\text{Null}(A) = \{v \mid Av = 0\}$

$= \{v \mid Rv = 0\}$

$= \text{Null}(R)$

To solve Null(R), we use non-pivot columns as parameter.

$\begin{cases} x_1 - 2x_2 + 3x_4 = 0 \\ x_3 - x_4 = 0 \\ x_5 = 0. \end{cases}$ $\begin{cases} x_2 = s \\ x_4 = t \end{cases} \Rightarrow \begin{cases} x_1 = 2s - 3t \\ x_2 = s \\ x_3 = t \\ x_4 = t \\ x_5 = 0. \end{cases}$

$\text{Null}(A) = \left\{ s \cdot \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -3 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \right\}$

Ans: $\left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \right\}$ forms a basis for Null(A).

Section 4.2 Rank



Let A be an $m \times n$ matrix.

number of pivot columns in RREF = number of nonzero rows in RREF

Definition

The **rank** of A is $\text{rank}(A) = \dim(\text{Col}(A)) = \dim(\text{Row}(A))$, the number of pivot columns in the RREF of A. A is of **full rank** if $\text{rank}(A) = \min\{m, n\}$.

Rank-Nullity Theorem

$\text{rank}(A) + \text{nullity}(A) = n$, the number of columns of A.

number of pivot columns in RREF + number of nonpivot columns in RREF

Theorem (full rank equals number of rows)

this implies $m = \min\{m, n\}$. or $m \leq n$.

The following are equivalent: (i) $\text{rank}(A) = m$; (ii) $\text{Col}(A) = \mathbb{R}^m$; (iii) the rows of A are linearly independent; (iv) $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b} \in \mathbb{R}^m$.

Question 2



Let A be an $m \times n$ matrix and B an $n \times p$ matrix. In this question we prove $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$.

1. Show that $\mathbb{R}^m$
   (a) $\text{Col}(A) \supseteq \text{Col}(AB)$;
   (b) $\text{Null}(AB) \supseteq \text{Null}(B)$.

   $\mathbb{R}^p$

Hint: a) $\text{Col}(A) = \text{span}(\text{colums in } A) = \{Av \mid v \in \mathbb{R}^n\}$

b) $\text{Null}(B) = \{v \mid Bv = 0\}$

2. Show that $\text{rank}(AB) \leq \text{rank}(A)$ using (a), and $\text{rank}(AB) \leq \text{rank}(B)$ using (b).

Question 2 — solution

a) Method 1

$A = \begin{pmatrix} | & & | \\ c_1 & \cdots & c_n \\ | & & | \end{pmatrix}$. $B = \begin{pmatrix} b_{11} & \cdots & b_{1p} \\ \vdots & & \vdots \\ b_{n1} & \cdots & b_{np} \end{pmatrix}$ $\quad \mathrm{col}(A) = \mathrm{span}(\{c_1, \ldots, c_n\})$.

$$AB = \begin{pmatrix} | & & | \\ c_1' & \cdots & c_p' \\ | & & | \end{pmatrix} \qquad \begin{pmatrix} | \\ c_i' \\ | \end{pmatrix} = b_{1i}\begin{pmatrix} | \\ c_1 \\ | \end{pmatrix} + b_{2i}\begin{pmatrix} | \\ c_2 \\ | \end{pmatrix} + \cdots + b_{ni}\begin{pmatrix} | \\ c_n \\ | \end{pmatrix} \in \mathrm{Span}(\{c_1, \ldots, c_n\})$$

In conclusion, this computation gives

$$\mathrm{col}(AB) = \mathrm{span}(\{c_1', \ldots, c_p'\}) \subseteq \mathrm{span}(\{c_1, \ldots, c_n\}) = \mathrm{col}(A).$$

Method 2

$$\mathrm{Col}(AB) = \{ \underbrace{AB}_{A}v \mid v \in \mathbb{R}^p \} \subseteq \mathrm{Col}(A) = \{ Au \mid u \in \mathbb{R}^n \}$$

$\{Bv \mid v \in \mathbb{R}^p\} \subseteq \mathbb{R}^n$.

Question 2 — solution

More formally, suppose that $w \in \text{col}(AB)$.
Then $\exists\, v \in \mathbb{R}^?$ s.t. $w = ABv = A(\underbrace{Bv}_{\in \mathbb{R}^n})$
so $w \in \text{col}(A)$. This implies that $\text{col}(AB) \subseteq \text{col}(A)$

b) $\text{Null}(AB) \supseteq \text{Null}(B)$

Suppose that $v \in \text{Null}(B)$. i.e. $Bv = 0$.

$\text{Null}(AB) \subseteq \text{Null}(B)$ is wrong in general.
eg. $A = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$. $B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

Then $(AB)v = A(Bv) = A0 = 0$. So $v \in \text{Null}(AB)$. We conclude that $\text{Null}(B) \subseteq \text{Null}(AB)$.

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2. a) $\text{rank}(AB) \leq \text{rank}(A)$
b) $\text{rank}(AB) \leq \text{rank}(B)$.

Hint: Use part 1 and the rank-nullity thm.

$A: m \times n$
$B: n \times p$
$AB: m \times p$.

a): $\text{rank}(AB) = \dim(\text{Col}(AB)) \leq \dim(\text{Col}(A)) = \text{rank}(A)$.

$\text{Col}(AB) \subseteq \text{col}(A)$.

b): $\text{Null}(AB) \supseteq \text{Null}(B) \Rightarrow \text{Nullity}(AB) \geq \text{Nullity}(B)$.

$\mathbb{R}^?$

$\text{rank}(AB) = p - \text{Nullity}(AB) \leq p - \text{Nullity}(B)$.

$\text{rank}(B)$

Question 3

Let A be an $m \times n$ matrix.

(a) Show that $\text{Null}(A^TA) = \text{Null}(A)$.

By Question 2 (b), it suffices to show

$\text{Null}(A^TA) \subseteq \text{Null}(A)$.

In other words, if $A^TAv = 0$, then we must show $Av = 0$.

Hint: A vector $u \in \mathbb{R}^n$ is zero $\Leftrightarrow u \cdot u = 0$.

(b) Suppose that A is of full rank and $m \geq n$. Show that A^TA is invertible, and that $(A^TA)^{-1}A^T$ is a left inverse of A.

$$\left[(A^TA)^{-1}A^T\right] \cdot A = (A^TA)^{-1}(A^TA) = I_n.$$

Question 3 — solution: (a)

a) As in the hint, we must show $A^TAv = 0$ then $Av = 0$.

$$Av = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix}.$$

$Av = 0 \iff Av \cdot Av = 0.$

$(Av)^T Av = v^T (A^T A v)$, and $A^TAv = 0$, so $= v^T 0 = 0$.

$Av \cdot Av = x_1^2 + x_2^2 + \cdots + x_m^2$.

So $x_1^2 + \cdots + x_m^2 = 0 \iff x_1 = x_2 = \cdots = x_m = 0$.

b) Suppose $m \geq n$ A is full rank. $\Rightarrow$ $\text{rank}(A) = n$ Then.

$\text{Nullity}(A^TA) = \text{Nullity}(A) = 0$.

rank–nullity thm: $\text{Nullity}(A) + \text{rank}(A) = n$.

Now A^TA is a square matrix with $\text{Nullity}(A^TA) = 0$.

Then $\{v \mid A^TAv = 0\} = \{0\}$.

$\iff A^TA$ is invertible. $\square$

Question 3 — solution: (b)

Next lesson

- **Recess Week**: Sat 19 – Sun 27 September. No class.
- Next lesson: **Tue 29 Sep**, **Sections 5.1 & 5.2 Orthogonality, and Orthogonal and Orthonormal Bases**. Please watch the video clips beforehand.